

2. Uber Rides Data Analysis using Python

Objective

The project analyzes Uber ride data to extract insights into ride patterns, such as peak hours, popular pickup/drop-off locations, and trends over time. The results aim to improve operational efficiency, customer experience, and strategic planning.

Key Analysis Goals

1. Identify peak demand hours and days.
2. Analyze pickup and drop-off locations to determine high-demand zones.
3. Examine trends in ride volumes over time (hourly, daily, or monthly).
4. Visualize insights to facilitate decision-making.

Tools and Technologies

- **Programming Language:** Python
- **Libraries:**
 - pandas and numpy for data preprocessing and manipulation.
 - matplotlib and seaborn for visualizations.
 - folium or geopandas for location-based visualizations (maps).
- **Dataset Format:** CSV file containing Uber ride details.